

9.1 Dams and its Classification

9.1.1 Dams

- Dam is a hydraulic structure constructed across a river in order to store water on its upstream side or divert water from river.
- Dam is an impervious or fairly impervious barrier put across a natural stream so that a reservoir is formed.
- This stored water is utilized when it is needed for various purposes.
- Due to construction of dam, a large area may get submerged on the upstream side.
- Dams are constructed mainly for irrigation or water power generation, diverting water into canals or other conveyances systems to the place of use, acting as a flood controlled reservoir.

Basis of Classification	Type	Common Examples
(a) Classification according to use	(i) Storage dam (ii) Diversion dam (iii) Detention dam	Gravity dam, Earth dam, Rockfill dam, Arch dam Weir, Barrage Dike, Debris dam
(b) Classification by hydraulic design	(i) Overflow dam (ii) Non overflow dam	Spillway Gravity dam, Earth dams, Rockfill dam
(c) Classification by materials	(i) Rigid dams (iii) Non rigid dams	Gravity dam, Arch dam, Buttress dam, Steel dam Earth dam, Rockfill dam

9.3 Gravity Dams

- Gravity dam is a masonry structure which increases stability against all applied loads by its weight alone (without depending on arch or beam action)
 - Gravity dam is a permanent type of dam.
 - Gravity dam is mostly straight in plan and approximately triangular in cross-section.
- ✓ **Low dams:** Dams upto 100 ft. (30.48 m) in height.
- ✓ **Moderate height dam:** Dams of heights between 100 ft. (30.48 m) and 300 ft. (91.44 m)
- ✓ **High dams:** Dams higher than 300 ft. (91.44 m)

9.8 Spillways

- A spillway is a water way provided in a dam (either through the dam, above the dam or somewhere remote from the dam) for carrying the surplus flood water from a reservoir safely to a downstream river.